Semester Wrap-Up and Thesis Progress

Barrier Function-Based Navigation for Precision Agriculture

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Agenda

1. Where We Started
2. What Was Achieved
3. What Is Next
4. Closing

End-of-Last-Semester Status

Foundation already in place:

- Literature review completed (CBFs, SLAM, greenhouse UAV applications).
- Experimental platform: Crazyflie 2.1 + OptiTrack + RTX workstation.
- Baseline PD control validated for hover and waypoints.
- MLP-based pose estimation explored (offline proof-of-concept).
- Published: *Energy-Aware Hybrid Event-Triggered Control for Micro-Drones* — IMAV 2025.

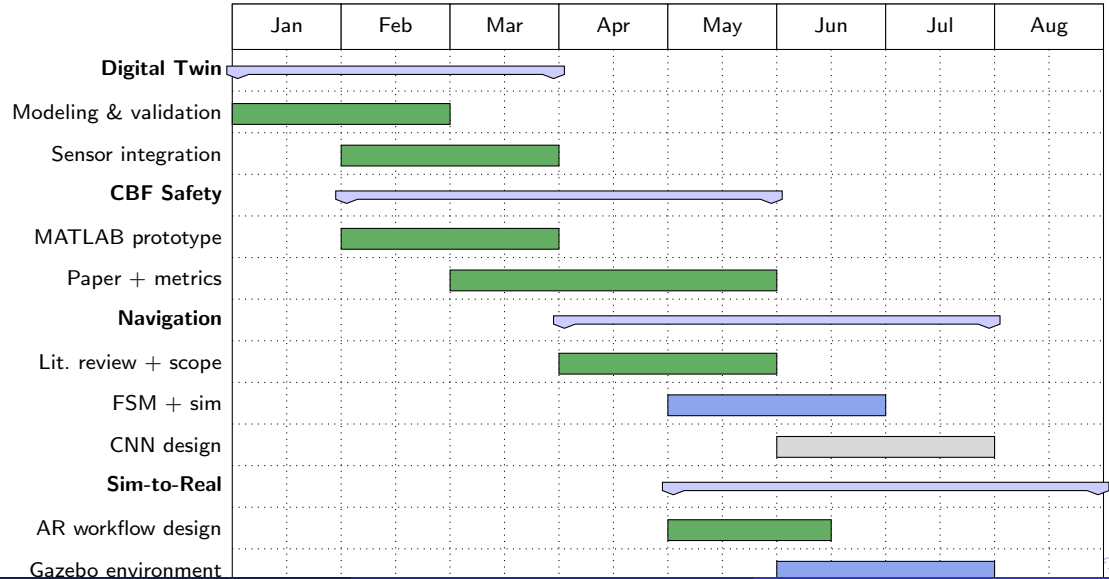
Open challenges:

1. No digital twin yet.
2. CBFs only conceptual, not implemented.
3. Navigation scope undefined — SLAM vs. topological?
4. No sim-to-real workflow.
5. Virtual environment not built.

Starting Point

Platform ready, but the **core thesis components** (digital twin, CBF safety, navigation

Planned Gantt for This Semester

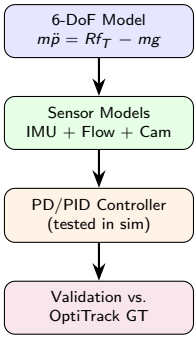


2.1 — Digital Twin: Creation & Validation

High-fidelity Crazyflie 2.1 simulation

- Full 6-DoF quadrotor dynamics model.
- Sensor suite: IMU, barometer, optical flow, monocular camera.
- Validated against **real flight data** (OptiTrack ground truth).
- Motor PWM distributions and attitude drift benchmarked.
- PD/PID low-level controllers tested in simulation first, then transferred to hardware.

Key result: Simulation-to-reality gap characterized and within acceptable bounds for controller design.



2.2 — CBF Safety Filter: MATLAB Prototype & Paper

3D Control Barrier Functions implemented

- Barrier functions $h(x)$ for obstacle/wall avoidance.
- CBF-QP optimization: minimal command modification for safety.
- Tested with PD baseline controller in Simulink.
- **5.7× speedup** measured via Simulink Profiler.
- DTW analysis with Bonferroni correction.
- GLM statistical validation of performance.

Published result: CBF manuscript finalized with full experimental metrics.

CBF-QP formulation:

$$u^* = \arg\min_u \|u - u_{\text{nom}}\|^2$$

$$\text{s.t.} \quad \dot{h}(x,u) + \alpha\, h(x) \geq 0$$

Guarantees:

- ✓ No wall collisions
- ✓ Safe human distances
- ✓ Altitude bounds
- ✓ Minimal control effort

2.3 — Navigation: Understanding Scope & Limitations

Literature-driven insight:
10 papers surveyed → convergent finding:

Full SLAM is unnecessary

For a greenhouse with *known structure* (parallel lanes), heavy SLAM is overkill. The literature converges on lightweight, topological approaches.

Chosen architecture:

1. **CNN** for lane centering + end-of-row detection.
2. **Topological FSM** for lane transitions.
3. **Boustrophedon planner** for complete coverage.
4. **CBF safety** wrapping all commands.

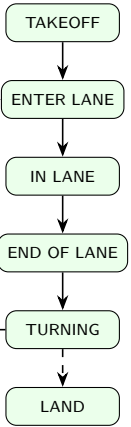
Key references:

- McGuire 2019: Bug alg. + flow → exploration without map.
- van Dijk 2024: Visual route following, $<20\,\mathrm{B/m}$ memory.
- Palossi 2019: DroNet on GAP8 @ 64 mW.

Beyond original scope:
This navigation architecture definition was *not originally planned* for this semester — it emerged from the literature review and represents an important scope clarification.

2.3 — FSM Simulation: Proof of Concept

Python simulation validating the topological FSM + simulated CNN perception

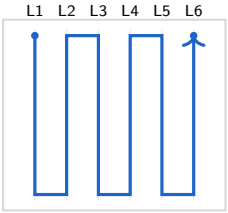


6-State FSM

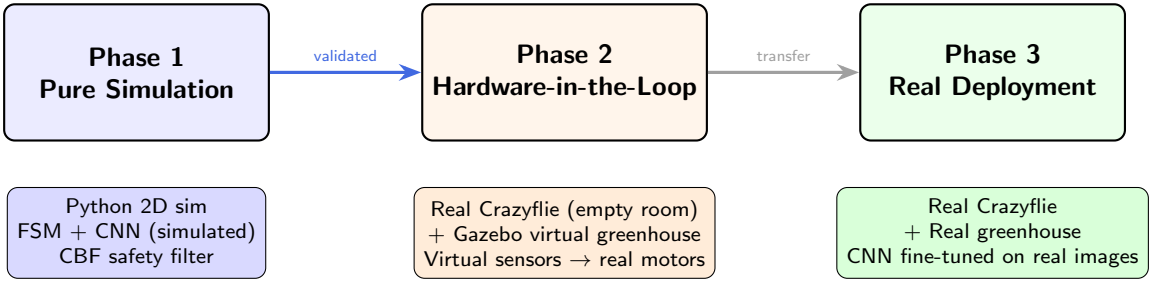
Simulation Metrics

Metric	Value
Lanes covered	6/6
Coverage	100 %
Flight time	153.9 s
Lateral RMSE	0.97 cm
FSM transitions	25
Time in-lane	78.8 %
Collisions	0

Trajectory (top-down)



2.4 — Augmented Reality & Sim-to-Real Workflow



Current status: Phase 1 complete, transitioning to Phase 2.
Gazebo environment under construction (replacing Unity).

Beyond original scope: This structured sim-to-real methodology was not in the original plan but provides a clear, reproducible path from simulation to hardware.

2.5 — Gantt Compliance Summary

Planned objectives — status:

Objective	Status
Digital twin creation	✓Done
Digital twin validation (vs. real)	✓Done
CBF prototyping (MATLAB)	✓Done
CBF paper metrics finalized	✓Done
Literature review update	✓Done
Extras (beyond plan):	
Navigation scope defined	✓Extra
FSM simulation built	✓Extra
Sim-to-real workflow designed	✓Extra

Assessment

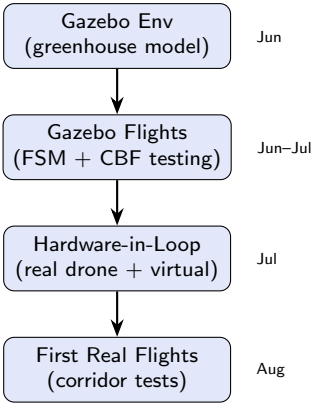
All planned semester objectives were accomplished on schedule. Additionally, work on **navigation scope** (2.3) and **sim-to-real workflow** (2.4) represents meaningful progress *ahead of the planned timeline*.

Publications this semester:

- IMAV 2025 (event-triggered control).
- CBF manuscript (under review).

Immediate Next Steps (Summer)

1. **Finish Gazebo virtual environment**
   - Greenhouse model with parallel crop rows.
   - Crazyflie URDF + sensor plugins (camera, flow, ToF).
   - ROS2 integration for FSM and CBF nodes.
2. **Test flights**
   - Validate FSM transitions in Gazebo.
   - CBF safety filter with simulated obstacles.
   - Lateral tracking accuracy under realistic noise.
3. **Sim-to-real start**
   - Hardware-in-the-loop: real drone + virtual greenhouse.
   - Identify and characterize the sim-to-real gap.



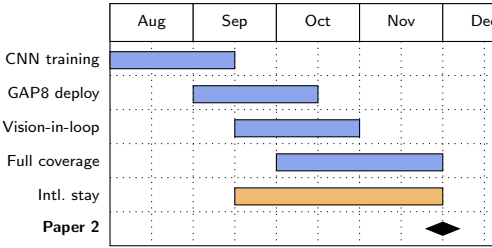
Medium-Term Goals (Next Semester)

Navigation & AI Design

- Train lane-centering CNN on synthetic greenhouse images.
- Quantize and deploy CNN on AI Deck (GAP8).
- Validate CNN inference rate ($\geq$10 fps) and power ($<$80 mW).
- Integrate CNN with FSM for vision-in-the-loop navigation.
- Full greenhouse coverage experiments.

International Research Stay

- Collaborative work at partner institution.
- Focus area: advanced perception or CBF extensions.



Thesis Contribution

Main Contribution

A **complete framework** for **safe, autonomous micro-UAV navigation** in structured agricultural environments (greenhouses), combining:

Lightweight Perception	**Formal Safety Guarantees**	**Complete Coverage**	**Validated Pipeline**
CNN on GAP8 (no SLAM)	3D CBF-QP (no collisions)	Topological FSM (no repeated lanes)	Sim-to-real (digital twin)

What makes this different:

- Runs entirely on a **27 g platform** — no external computation, no GPS, no infrastructure beyond the drone itself.
- **Provable safety** via Control Barrier Functions, not just heuristic collision avoidance.
- **Full environment coverage** guaranteed by topological planning, not random exploration.
- End-to-end **reproducible methodology**: digital twin → simulation →

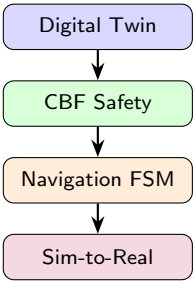
Take-Home Message

This semester established all core thesis components:

1. **Digital twin** — validated against real hardware.
2. **CBF safety** — prototyped, benchmarked, paper submitted.
3. **Navigation architecture** — scope clarified, FSM simulated.
4. **Sim-to-real workflow** — 3-phase pipeline designed.

Next milestone: A Crazyflie flying through a simulated greenhouse corridor with real-time FSM + CBF safety — ready by end of summer.

Thank you for your attention.



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